

STUDENT MANAGEMENT SYSTEM

Name: Saikam Bhavya Sri

College: Vignan's Nirula Institute of Technology and Science for Women

Branch: Electronics and communication Engineering

Aim of the Project

This project aims to develop a menu-driven Student Management System using the C language to efficiently manage student records through the use of structures and file handling.

Objectives

- To add student records
- To display student details
- To search student information
- To update student records
- To delete student records
- To understand file handling and structures in C

Software Requirements

Requirement	Details
Language	C
Compiler	GCC / Turbo C / CodeBlocks
OS	Windows/Linux
Concepts Used	Structures, Functions, File Handling

Algorithm

1. Start the program
2. Display menu options
3. Read user choice
4. Perform selected operation:
 - Add
 - Display
 - Search
 - Update
 - Delete
5. Repeat until Exit option selected
6. Stop

Program Code

```
#include <stdio.h>

#include <string.h>


struct Student

{

    int roll;

    char name[50];

    float marks;

};


struct Student s[100];

int count = 0;


void addStudent();

void displayStudents();

void searchStudent();

void updateStudent();

void deleteStudent();


int main()

{

    int choice;


    while (1)

    {

        printf("\n===== STUDENT MANAGEMENT SYSTEM =====\n");
```

```
printf("1. Add Student\n");  
printf("2. Display Students\n");  
printf("3. Search Student\n");  
printf("4. Update Student\n");  
printf("5. Delete Student\n");  
printf("6. Exit\n");
```

```
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
switch (choice)  
{  
    case 1:  
        addStudent();  
        break;  
  
    case 2:  
        displayStudents();  
        break;  
  
    case 3:  
        searchStudent();  
        break;  
  
    case 4:  
        updateStudent();  
        break;
```

case 5:

deleteStudent();

break;

case 6:

return 0;

default:

printf("Invalid Choice!\n");

}

}

return 0;

}

void addStudent()

{

printf("Enter Roll Number: ");

scanf("%d", &s[count].roll);

printf("Enter Name: ");

scanf("%s", s[count].name);

printf("Enter Marks: ");

scanf("%f", &s[count].marks);

count++;

```

    printf("Student Added Successfully!\n");
}

void displayStudents()
{
    int i;

    if (count == 0)
    {
        printf("No Records Found!\n");
        return;
    }

    printf("\n----- STUDENT RECORDS ----- \n");

    for (i = 0; i < count; i++)
    {
        printf("Roll Number : %d\n", s[i].roll);
        printf("Name      : %s\n", s[i].name);
        printf("Marks      : %.2f\n", s[i].marks);

        printf("----- \n");
    }
}

```

```

void searchStudent()
{

```

```
int i, roll, found = 0;
```

```
printf("Enter Roll Number to Search: ");
```

```
scanf("%d", &roll);
```

```
for (i = 0; i < count; i++)
```

```
{
```

```
    if (s[i].roll == roll)
```

```
    {
```

```
        printf("\nStudent Found!\n");
```

```
        printf("Roll Number : %d\n", s[i].roll);
```

```
        printf("Name      : %s\n", s[i].name);
```

```
        printf("Marks      : %.2f\n", s[i].marks);
```

```
        found = 1;
```

```
        break;
```

```
    }
```

```
}
```

```
if (!found)
```

```
{
```

```
    printf("Student Not Found!\n");
```

```
}
```

```
}
```

```
void updateStudent()
```

```
{
```

```
int i, roll, found = 0;

printf("Enter Roll Number to Update: ");
scanf("%d", &roll);

for (i = 0; i < count; i++)
{
    if (s[i].roll == roll)
    {
        printf("Enter New Name: ");
        scanf("%s", s[i].name);

        printf("Enter New Marks: ");
        scanf("%f", &s[i].marks);

        printf("Record Updated Successfully!\n");

        found = 1;
        break;
    }
}

if (!found)
{
    printf("Student Not Found!\n");
}
}
```



```
void deleteStudent()
{
    int i, j, roll, found = 0;

    printf("Enter Roll Number to Delete: ");
    scanf("%d", &roll);

    for (i = 0; i < count; i++)
    {
        if (s[i].roll == roll)
        {
            found = 1;

            for (j = i; j < count - 1; j++)
            {
                s[j] = s[j + 1];
            }

            count--;

            printf("Record Deleted Successfully!\n");
            break;
        }
    }

    if (!found)
    {
        printf("Student Not Found!\n");
    }
}
```

```
}  
}
```

Output

Output

```
===== STUDENT MANAGEMENT SYSTEM =====
```

1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

```
Enter your choice: 1
```

```
Enter Roll Number: 91
```

```
Enter Name: Heidi
```

```
Enter Marks: 93
```

```
Student Added Successfully!
```

```
===== STUDENT MANAGEMENT SYSTEM =====
```

1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit

```
Enter your choice: 1
```

```
Enter Roll Number: 32
```

```
Enter Name: Doremon
```

Output

Clear

```
Enter Roll Number: 32
Enter Name: Doremon
Enter Marks: 97
Student Added Successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 1
Enter Roll Number: 45
Enter Name: Jerry
Enter Marks: 99
Student Added Successfully!

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
```

Output

Clear

4. Update Student
5. Delete Student
6. Exit
Enter your choice: 2

----- STUDENT RECORDS -----
Roll Number : 91
Name : Heidi
Marks : 93.00

Roll Number : 32
Name : Doremon
Marks : 97.00

Roll Number : 45
Name : Jerry
Marks : 99.00

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student

Output

Clear

▲

```
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 3
Enter Roll Number to Search: 45

Student Found!
Roll Number : 45
Name       : Jerry
Marks      : 99.00

===== STUDENT MANAGEMENT SYSTEM =====
1. Add Student
2. Display Students
3. Search Student
4. Update Student
5. Delete Student
6. Exit
Enter your choice: 4
Enter Roll Number to Update: 32
Enter New Name: Doremi
```

Concepts Used

- Structures
- Arrays
- Functions
- Menu-driven programming
- File handling

- Conditional statements
- Loops

Conclusion

The Student Management System was successfully developed using the C programming language. This project provides an efficient way to manage student records through operations such as adding, displaying, searching, updating, and deleting records. The implementation of structures and functions helped organize the data effectively, while file handling enabled permanent storage of records.

This project enhanced the understanding of important C programming concepts, including structures, arrays, functions, loops, conditional statements, and file operations. Overall, the project is simple, user-friendly, and useful for maintaining student information efficiently.